This research report compares three differing explanations of the dynamic interrelationships between internal and external innovation-related communication in a new organizational form. In the functional specialization explanation, individuals are said to focus on the mix of internal and/or external communication dictated by their formal positions. The communication stars explanation suggests that individuals maintain similar levels of communication in both networks. The cyclical model posits a more dynamic pattern that shifts back and forth between internal and external communication, depending on the consequences of their prior communication behavior. The new organizational form examined for three years was the Cancer Information Service, a geographically dispersed federal government health information program. Our results indicated that there was a lagged effect for the communication stars explanation.

Keywords: Boundary Spanning, Communication Stars, Health Information, Innovation, Networks

Internal and External Communication, Boundary Spanning, and Innovation Adoption: An Over-Time Comparison of Three Explanations of Internal and External Innovation Communication in a New Organizational Form

J. David Johnson
University of Kentucky, Lexington

Hui-Jung Chang
National Sun Yat-sen University, Kaohsiung, Taiwan

External communication links, which are often associated with boundary spanning, are critical to enhancing innovations since they provide opportunities for learning and for securing needed resources (Goes & Park, 1997) and for the diffusion of ideas between and within organizations (Cziepel, 1975; Daft, 1978; Ghosal & Bartlett, 1987: Kimberly, 1978; Robertson & Wind, 1983). Such links are the mechanism that operationalizes

J. David Johnson (Ph.D., Michigan State University, 1978) is Professor and Dean of the College of Communications and Information Studies at the University of Kentucky. His current research focuses on organizational communication structure, innovation, and information seeking.

Hui-Jung Chang (Ph.D., Michigan State University, 1996) is Assistant Professor of the Institute of Communications Management at the National Sun Yat-sen University in Taiwan. Her current research focuses on boundary-spanning communication, organizational media usages, media management, and public relations.

A prior version of this paper was presented to the Organizational Communication Division at the International Communication Association Annual Convention, Montreal, Canada, 1997. Part of this paper was based on the dissertation

The Journal of Business Communication, Volume 37, Number 3, July 2000, pages 238-263

environmental cues to the internal organizational structure (Corwin, 1972; Lozada & Calantone, 1996; Spekman, 1979). The present study examines external communication longitudinally in the more voluntary communication environment of a new organizational form. Most of the prior literature on innovaton-related communication has emphasized the constraints posed by a person's formal position; however, more recently it has been suggested that new, emerging designs provide opportunities for individuals to shape their own innovation related communication patterns.

We will directly contrast explanations derived from formally prescribed (functional specialization) and emergent (communication stars) theoretical positions (Monge & Eisenberg, 1987; Johnson, 1993). In the functional specialization model, individuals are predicted to focus on either internal or external communication depending on their formal functional positions. The communication stars explanation argues that individuals are disposed to the same levels of communication in both internal and external networks. Yet a third model offers a cyclical explanation, positing that individuals rotate their internal and external communication in a dynamic pattern depending on organizational requirements.

Most of the current organizational literature tends to favor virtual designs. Virtual designs are based on market assumptions (Galbraith, 1995) and place an increasing burden on individuals to find their way amidst chaos (Miles, Snow, Matthews, Miles, & Coleman, 1997). However, other studies have suggested that the increasing complexity of these forms needs to be balanced by a concomitant interest in formalization (Johnson, LaFrance, Meyer, Speyer, & Cox, 1998; Johnson, Meyer, Berkowitz, Ethington, & Miller, 1997), which reduces uncertainties arising in these new forms. A host of environmental factors contribute to the development of new organizational forms: concerns about personnel costs (e.g., pensions, health costs); external pressures to keep the number of

from the second author under the direction of the first author. The research was supported by Subcontract No. 737-4241 from the AMC Cancer Research Center for P01 CA57586-01A1 grant from the National Cancer Institute. The conclusions reached in this paper are solely those of the authors and do not necessarily reflect the views of the National Cancer Institute or the AMC Cancer Research Center.

We would like to thank the Office of Cancer Communications as well as the CIS offices which participated in the data collections. We would like to thank Deb Tigner for her help in preparing and mailing the questionnaires. We would also like to extend thanks to the other members of the Network Analysis Advisory Board for their help throughout the many phases of the research process: Donna Cox, Jo Beth Speyer, Maureen McClatchey, and Diane Ruesch. We would also like to thank the other members of the Network Analysis Team: Caroline Ethington, Toru Kiyomiya, Betty LaFrance, and Marcy Meyer

Correspondence concerning this article should be addressed to Hui-Jung Chang at: Institute of Communications Management, College of Management, National Sun Yat-sen University, Kaohsiung, 80424, Taiwan or <changhj@cm.nsysu.edu.tw>.

members on their permanent staff low; uncertainty reduction; needs to pool knowledge and information or to create it in the case of research and development (R&D) firms (Gibson & Rogers, 1994); increasing access to information by reducing institutional barriers (DeBresson & Amesse, 1991); affiliation (e.g., with a more credible national organization); and building mutually supportive power bases to lobby various stakeholders. Fundamentally, consortiums are formed so that their members can accomplish more than they could do on their own. Increasingly, organizations find that they are either strapped for resources or are pursuing such large projects that they must pool their resources to pursue innovations (Browning, Beyer, & Shelter, 1995; DeBresson & Amesse, 1991; Hakansson & Sharma, 1996).

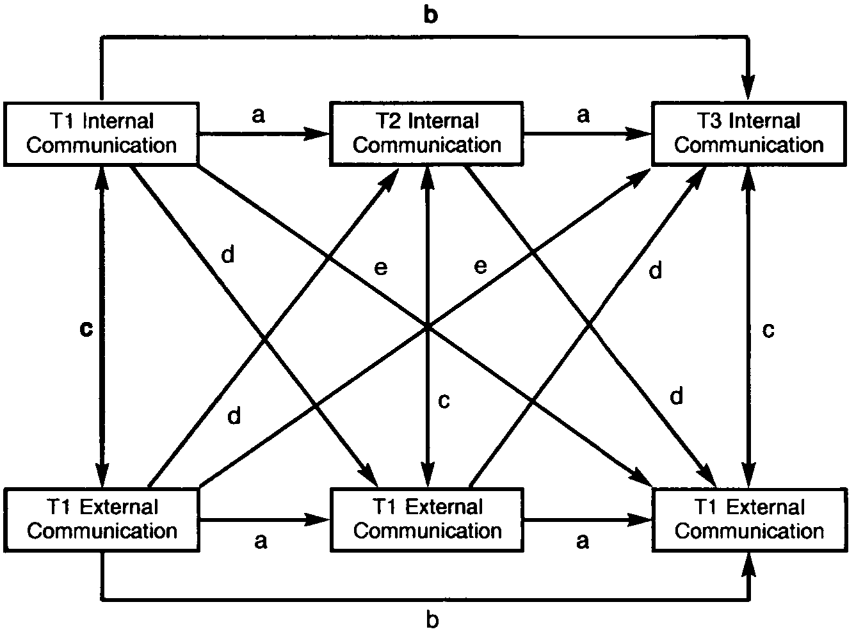
Creating new organizational forms is difficult, particularly in health care (Arnold & Hink, 1968; Farace, Monge, Bettinghaus, Eisenberg, White, Kurchner-Hawkins, & Williams, 1982; Kaluzny, Lacey, Warnecke, Hynes, Morrissey, Ford, & Sondik, 1993; Kaluzny & Warnecke, 1996; Luke, Begun, & Pointer, 1989). At least five barriers have been identified: (a) Cooperating agencies often have different missions (e.g., providing social support vs. treatment for cancer patients); (b) outcome and effectiveness measures differ among agencies; (c) the coordination costs are too heavy (DeBresson & Amesse, 1991) to truly integrate the efforts of diverse organizations (Arnold & Hink, 1968); (d) members of coalitions may have multiple goals (Stevenson, Pearce, & Porter, 1985), and may resent the loss of decision-making latitude; and (e) the cost of managing their linkages increases (Oliver, 1990). In the face of these obstacles, there is an increasing need to develop new theories and fresh perspectives based on empirical data about the operation of these new organizational forms (Luke et al., 1989). Indeed, the ability of a society to create new organizational forms may directly affect its ability to adapt to new environmental circumstances (Romanelli, 1991), such as an increasingly competitive global environment.

The success of new organizational forms depends on managing interorganizational relationships through external communication. Consequently, more empirical study is necessary to determine the ideal balance between formalized structure and emergent communication networks in these new organizational forms. Specifically, research must give more attention to individual patterns of internal and external communication. Research is particularly important now given the increased attention paid to new organizational forms and an increased recognition that within these new forms there are not clear boundaries, but rather gradations of affiliation between entities (Sheppard & Tuchinsky, 1996).

Three Explanations of the Interrelationships Between Internal and External Communication

Figure 1 contains a classic panel representation of the interrelationships

Figure 1.
Classic Panel Paths for Differing Explanations of the Interrelationships between Internal and External Communication



T1, T2, and T3 = data-gathering periods

a = Stability at consecutive time points
b = Stability at lagged time points
c = Cross-sectional interrelationships
d = Consecutive interrelationships
e = Lagged interrelationships

between internal and external communication over three points in time (see Finkel, 1995; Williams & Podsakoff, 1989). In this research report we will examine possible configurations of relationships, or paths in a classic path analytic sense, for three contrasting models: functional specialization, communication stars, and cyclical. Each explanation predicts different key paths in Figure 1, where *a* and *b* paths represent the stability of internal or external communication at consecutive or lagged periods respectively, *c* paths represent the cross-sectional interrelationships between internal and external communication at any one time, *d* paths specify relationships between internal and external communication across consecutive points in time, and *e* paths represent the lagged interrelationships between internal and external innovation communication from Time 1 to 3.

Functional Specialization

Since organizations must adapt to their environments, a number of formal structures and associated functional roles are created explicitly to deal with them (Galbraith, 1974). For example, boundary spanners (e.g., department heads, customer service representatives) maintain external communication contacts because of their formally assigned roles (At-Twaijri & Montanari, 1987; Burk, 1994; Friedman & Podolny, 1992; Grover, Jeong, Kettinger, & Lee, 1993; Keller, Szilágyi, & Holland, 1976; Lysonski & Johnson, 1983; Schwab, Ungson, & Brown, 1985; Singh, Goolsby, & Rhoads, 1994; Stevenson, 1990). Boundary spanners are responsible for making communication contacts with external information sources and supplying their colleagues with information concerning the outside environment, all while maintaining an organization's autonomy (Adams, 1976; Aldrich & Herker, 1977).

Boundary spanners play an important role in the diffusion of ideas between and within organizations (Albrecht & Ropp, 1984; Cziepel, 1975; Daft, 1978; Ghosal & Bartlett, 1987; Schwab, Ungson, & Brown, 1985). Nowhere is this more true than in the health care environment (Robertson & Wind, 1983). Boundary spanners are the mechanism that operationalizes environmental cues to the internal organizational structure (Jemison, 1984; Lozada & Calantine, 1996; Spekman, 1979). A substantial proportion of the boundary spanning literature has implicitly adopted a two-step communication process (e.g., Katz & Lazarsfeld, 1955), with an emphasis on information flowing through boundary spanners who act as opinion leaders in their organizations. However, in new organizational forms most individuals engage in some boundary spanning behavior, rendering more traditional organizational boundaries increasingly arbitrary (Starbuck, 1976). While the literature suggests various types of boundary spanning communication activities, few studies simultaneously have examined internal (between organizational units) and external (with other organizations) communication patterns over time, especially in relation to innovation processes (Goes & Park, 1997). Inherent in functional specialization is that individuals will concentrate on either internal or external communication, depending on their formal position. A more structured way to represent the functional specialization approach is as follows:

H1: A functional specialization model posits

A: Positive relationships between internal communication at all three times (the *a* and *b* paths in Figure 1).

B. Positive relationships between external communication at all three times (the *a* and *b* paths in Figure 1).

C. Negative relationships between internal and external communication at each time (*c* paths) and in their cross-lagged relationships (*d* and *e* paths).

Communication Stars

Disputing the functional specialization explanation, some boundary spanning literature suggests that the two distinctive external and internal communication roles can be played by the *same* person (Aldrich & Herker, 1978; Allen, 1989; Friedman & Podolny, 1992; Katz & Tushman, 1981; Tushman & Scanlan, 1981a, 1981b). Consequently, research has also focused on boundary spanners who communicate externally as well as internally. For example, Nagpaul and Pruthi (1979) reported that technical R&D gatekeepers used external contacts for idea-generation and internal networks for problem solving. Similar findings are reported in Tushman and Scanlan's (1981b) investigation in a high-tech R&D facility. These researchers found that boundary spanners were likely to be identified as a valuable internal source of new information. Thus, when a person's communication spans both internal and external networks, it would appear that the two networks are mutually reinforcing. Some management research also recognizes boundary spanning activities both inside and outside the organization (Mintzberg, 1973). For instance, middle management sometimes requires individuals to be both internal and external stars. Recent research has suggested that balancing internal and external ties establishes individual influence (Manev & Stevenson, 1996).

While it would seem obvious that there are finite limits to the amount of communication in which one can engage (Baker, 1992), several studies suggest that individuals who are high communicators in one setting are also high in others. That is, heavy users of one information medium related to work are likely to be users of other media that also carry this same information (Blau & Alba, 1982; Caroll & Teo, 1996; Paisley, 1980; Weedman, 1992), which is also a finding of more general media use studies (Berelson & Steiner, 1964).

Based on the preceding discussion, it seems reasonable to suggest that boundary spanners focus on both internal and external activities simultaneously. Moreover, boundary spanners acquire relevant information from their extensive external contacts and filter and feed the information into the organization. Boundary spanners also often seek out information from each other, and peers tend to rate them as influential (Paisley, 1980; Reynolds & Johnson, 1982). A more structured way to represent the communication stars approach is our second hypothesis:

H2: A communication star model posits

A: Positive relationships between internal communication at all three points in time (the *a* and *b* paths in Figure 1).

B. Positive relationships between external communication at all three times (*a* and *b*).

C. Positive relationships between internal and external communication at each point in time (*c* paths) and in their cross-lagged relationships (*d* and *e* paths).

Cyclical Model

A melding of both of the functional specialization and communication stars literatures suggests a third model which accounts for the interrelationships between internal and external communication patterns. This alternative model suggests that communication relationships may shift due to the systemic consequences resulting from the boundary spanning activities and dynamic organizational requirements. For instance, to avoid role conflict, boundary spanners might focus their efforts in one network (internal or external). As the R&D literature suggests, importing external ideas might result in considerable internal communication generating internal innovations, which, in turn, are then exported to other organizations through external communication. A more structured way to represent the cyclical approach is our third hypothesis:

H3: A cyclical model posits

A. Negative relationships between internal communication at consecutive time points (the *a* paths in Figure 1).
B. Positive relationships between internal communication at Time 1 and Time 3 (the *b* paths).
C. Negative relationships between external communication at consecutive time points (the *a* paths).
D. Positive relationships between external communication at Time 1 and Time 3 (the *b* paths).
E. Negative relationships between internal and external communication at each point in time (*c* paths) and in their lagged relationships (*e* paths).
F. Positive relationships between internal and external communication at consecutive points in time (*d* paths)

Methods

The present study evaluated the three competing boundary spanning explanations in a unique setting, focusing on innovation-related communication in a new organizational form. In this section we describe the unique features of the organization we studied (Cancer Information Service). We then focus on the approach adopted to define the boundaries of the organization, the participants in the research studies, and the specific questions asked of the study participants.

The Cancer Information Service as a New Organizational Form

This study examines a confederation of contractors who provided services to the Cancer Information Service (CIS). The CIS was established in 1975 by the National Cancer Institute (NCI) to disseminate accurate, up-to-date information about cancer to cancer patients, their relatives and friends, health care professionals, and the general public (Morra, Bettinghaus, & Marcus, 1993; Morra, Van Nevel, Nealon, Mazan, & Thomsen, 1993). In response to this mandate, the CIS currently maintains a network of 19 Regional Offices (ROs) that are typically linked to NCI-funded regional cancer centers. The activities of the CIS network are coordinated and supervised by the Office of Cancer Communications (OCC) at the NCI. These activities fall into two broad categories: (a) responding to requests for information over the telephone (the CIS operates a toll-free telephone number in which callers are automatically triaged to their regional office for response from a trained and certified Cancer Information Specialist), and (b) conducting community outreach activities. The outreach program of the CIS serves as a catalyst and focal point for cancer education at the state and regional level and is the focus of much of the external communication examined here.

The ROs are brought together by a classic fee-for-services contract. In effect, the contract hires temporary organizations for five years to conduct a specified scope of work for the NCI. The unique characteristics of the CIS become apparent when contrasted with more conventional organizational forms. For instance, although the ROs are formally members of other organizations, the agency itself has many of the characteristics of unitary organizations, such as centrally determined goals, a formal bureaucratic structure of authority, a division of labor, formal plans for coordination (e.g., sharing calls), a high normative commitment to providing service to callers, and targeted outreach activities to priority audiences. Performance standards are set nationally and are monitored by an extensive formal evaluation effort (Kessler, Fintro, Muha, Wun, Annett, & Mazen, 1993). However, important personnel issues such as salaries and fringe benefits are determined at the RO level. Thus, regions internally boundary span with each other and OCC, while maintaining external communication contacts with other organizations in their communities. External communication primarily developed community-based initiatives (e.g., breast cancer awareness month), while internal communication focused on implementing new techniques (e.g., making outcalls to encourage mammography screening) for reaching the public.

Boundaries of the CIS

For the purpose of this research project, the composition of the CIS internal network was based on nominalist views of network boundaries (Lauman, Marsden, & Prensky, 1983). From a nominalist perspective, five major functional roles, representing key decision-makers within the CIS, were examined: (a) Office of Cancer Communications (OCCs) staff at NCI and (b) Project Directors (PDs), (c) Outreach Managers (OMs), (d) Telephone Service Managers (TSMs), and (e) Principal Investigators (PIs) at ROs. OCCs are in charge of coordinating and supervising the activities of the regional CIS network. PDs engage in a mixture of internal and external communication coordinating work with OCCs, other ROs, or their local cancer centers. TSMs primarily focus on the internal telephone communication and referral services. OMs are active in the external network since they are responsible for developing relationships with community organizations. PIs are the principal investigators of the CIS contract. Thus, we examined formal roles with a mix of internal and external communication responsibilities.

Sampling Interval

This study was part of a larger project designed to evaluate the impact of three planned innovations over four years (see Johnson, Berkowitz, Ethington, & Meyer, 1994a; Johnson, Bettinghaus, Woodworth, Fleisher, Ward, & Meyer, 1997; Meyer, Johnson, & Ethington, 1997). Consequently, selection of the sampling interval was particularly critical given the complexity of the overall investigation. As a result of extensive pretesting and discussions with members of the network about internal communication reports, we decided to focus on a three-day period every three months. In addition, data were collected by rotating days of the week and weeks of the month throughout the duration of the project. Internal communication network data were regularly collected at each of 14 scheduled sampling periods (see Johnson et al., 1994a). Unlike the internal communication data gathered at relatively frequent intervals, external communication data (i.e., radial communication network data representing individual reports of their linkages to other organizations) were collected once a year for three years. The sampling was frequent enough to detect major cycles of activities within the CIS system, while recognizing limitations of respondent memory and the vast volumes of data that might be generated.

Data Collection

Communication data on external contacts were collected at three points of time: T1, T2, and T3 (c.f., Johnson, Chang, Ethington, Meyer, & LaFrance, 1994; Johnson, Chang, La France, et al., 1996a, b).[1] At each data collection period, a package was sent to respondents containing a communication log and a battery of questions relating to their external communication contacts. To help ensure completion, the self-report questionnaires were mailed to the respondents approximately 10 days before the sample time period. A personalized letter explained the issues that would be examined and urged participation in the project. In addition, participants also received an e-mail to notify them that they would soon receive the questionnaires. A second e-mail was sent the day before the sample time period, reminding participants that they should begin recording their communication contacts for the next three days. A third e-mail was sent the day after the sample time period concluded, reminding participants to return their questionnaires in the provided stamped, self-addressed envelope. In fact, many follow-up steps (e.g., letters, faxes, e-mails) recommended by the literature (e.g., Dillman, 1978, 1991) were used in these recurring data collections. Perhaps because of the extensive follow-up efforts, we achieved a satisfactory response rate of 93 percent, 93 percent, and 95 percent at T1, T2, and T3, respectively.

Respondents and Level of Analysis

The sample sizes were 110, 103, 121 at T1, T2, and T3, respectively, with the core cohort remaining essentially stable. Study participants were highly educated: 92 percent of the respondents had earned college degrees, 51 percent of which were graduate degrees. Interestingly, fewer than one-third of respondents had worked for the CIS for five years or more.

Many prior studies have used organizations as the unit of analysis, distinguishing between internal and external communication on the level of intra-organization and inter-organization respectively (Allen, 1989; Zoch, 1993). Yet, others have used alternative levels of analysis. For example, Tushman and Scanlan (1981a) chose the department as the unit of analysis in an R&D setting. These researchers defined internal communication as the communication occurring within the department, while external communication reflected activities taking place on an intra-organizational and extra-organizational level. As a result, external activities in one study may be treated as internal communication in others, and internal communication stars on an intra-organizational level may be defined as external communication stars on a group level.

CIS is the unit of analysis in the present study. Thus, internal communication refers to the communication occurring among and between the 19 ROs and OCC, while external communication denotes the communication contacts that occurred with the organizations outside the CIS network (e.g., American Cancer Society, Health Department, and so on). This study focused on the boundary spanning communication of all individuals, given the assumption that communication is widely distributed in new organizational forms.

Internal Communication

Respondents were asked to record the interpersonal communication contacts they initiated or received within CIS network for a three-day period.[2] They were instructed to record the inter-regional communication on the national level.[3] For the respondents' convenience, a directory of individuals within the CIS network and pre-dated pages of the log were provided. Respondents were asked to record their innovation-related communication concerning intervention strategies within the network. These contacts included initiatives that related to the development or implementation of programs which focused on reaching various target populations, such as counseling protocols for special target populations, targeted outreach activities using the telephone, and responses to calls associated with communication campaigns (see Meyer et al., 1997 for more detail).[4]

External Communication

Respondents were asked to record or estimate the number of times they initiated or received contacts with a member representing outside organizations about intervention strategies. Respondents were also instructed to count each individual only once. For respondents' convenience and for reliability purposes, participants were also provided with an operational definition of intervention strategies. The list of the outside organizations was developed after considerable collaboration with the CIS staff, and the list was finalized after several pretests within the CIS network (see Johnson et al., 1994a). Separate questionnaires were developed for OCCs and other functional groups because of their job requirements.

Results

In this section of the report we first provide descriptive data, followed by a description of the analysis plan. We then compare each of the three models used to study the CIS.

Descriptive Statistics

The means, standard deviations, ranges, and censored correlations for the variables are presented in Table 1. For external communication, approximately one-sixth of respondents reported zero contacts across the three time points (n = 22, 21, & 27, respectively). Over one half of the respondents reported having 0 contacts for internal communication at T1 and T2 (n = 71 and 64 respectively) and even more reported no contact at T3 (n=98). These data led us to adopt a censored variable approach, normalizing the variables at the lower ranges of observed values. The data were subjected to LISREL analysis by means of the PRELIS computer software (see SPSS, 1993).

In general, respondents had considerably more external than internal contacts. Comparable numbers of communication contacts were observed for internal communication at T1 and T2 (mean = .65 and .68, respectively), and for external communication at T2 and T3 (mean = 10.97 and 10.82, respectively). Standard deviations for both internal and external communication across all three time points were relatively high (ranging from .60 to 1.17 and from 13.45 to 14.68, respectively). Extreme outliers (those with scores above 80 for external communication) were removed

Table 1.
Means, Standard Deviations, Ranges, and Censored Correlations over Three Points in Time

Variables	Mean	S.D.	Maximum	Minimum	T1in	T2in	T3in	T1ex	T2ex	T3ex
T1internal	.65	1.16	7	0	–					
T2internal	.68	1.17	6	0	.37	–				
T3internal	.25	.60	4	0	.18	.30	–			
T1external	12.45	14.19	72	0	.34	.30	.03	–		
T2external	10.97	14.68	62	0	.29	.17	.06	.29	–	
T3external	10.82	13.44	77	0	.22	.18	.02	.40	.45	–

before censored correlation coefficients were calculated for the model. The highest correlation was observed between external contacts at T2 and T3 r = .45), while the lowest correlation was observed between internal communication and external communication at T3 r = .02). There were no obvious problems with multicollinearity present in the censored correlation matrix.

Analysis Plan

Analysis of panel data has been a source of controversy because of the intractable difficulties that a researcher often confronts, such as an inability to account for random measurement error, correlated disturbances attributable to unspecified third variables, and identification problems (see Williams & Podsakoff, 1989, for an exhaustive discussion). In this research we used an iterative approach for estimating an optimal model from a nested series of models for evaluating structural parameter estimates by means of LISREL. Sometimes referred to as path analysis, LISREL is a general analytic technique for estimating a linear structural equation system. One of the unique advantages of LISREL is that it provides estimates of fit for the entire model to the data. Consequently, LISREL ameliorates the problems inherent in a classic approach to analysis of panel models (Finkel, 1995; Williams & Podsakoff, 1989). The approach also permits the analysis of residuals, reciprocal relationships, and the equality of causal parameters over-time (Finkel, 1995; Williams & Podsakoff, 1989).

To determine superior parameter estimates a series of nested models using the Normed Fit Index (NFI), which is (Fn – Fm)/Fn, where Fn = χ^2/n for the null model and Fm =χ^2/n for the model under examination, were evaluated. NFI can range from 0 to 1.00, is independent of sample size, and reflects the goodness of fit of competing models (Bentler & Bonett, 1980; Williams & Podsakoff, 1989). Model comparisons can be used to assess the adequacy of causal lags and the relative importance of different variable groups (Williams & Podsakoff, 1989). The question of fit is becoming increasingly complicated (see Bollen & Long, 1993). We therefore also report the Akaike Information Criterion (AIC) because it differs from the NFI in several important ways: It does not range from 0-1, it does not apply just to nested models, and it favors simple models (Tanaka, 1993). The AIC was the test statistic, (chi-square), + 2 times the number of free or estimated parameters.

Following the iterative procedure outlined by Finkel (1995), we tested a number of models. In the null model (see Table 2) none of the parameters were estimated, except that the on diagonal elements of psi were constrained to equal the same value of one (consult Anderson & Gerbing, 1988). This model became the baseline to which subsequent models were nested. The model yielded a chi-square value of 65.01 with 20 *df.* In model B through H, differing combinations of substantive (beta) paths, lagged paths over two points in time, psi, or covariances of endogenous variables, and theta epsilon, or measurement error, paths were estimated and/or constrained to equal each other. Given roughly equal time lags, similar paths during different time intervals should be equal. Hence, the rough equivalence of these paths would be one indication of the appropriateness of these causal intervals (Finkel, 1995). In model G, for example, paths between T1 and T3 were estimated, the mirror gamma paths and psi correlations at T2 and T3 were constrained to equal each other (e.g., between internal communication at T1 and external communication at T2 and between internal communication at T2 and external communication at T3), and the diagonal elements of psi were constrained to equal each other. Model G had a chi-square of 5.99 with 4 *df,* a NFI of .91, and an AIC of 39.99.

Models A through H exhibited a narrow range of scores on both the AIC and NFI. Model H, with a score of .91, had the superior NFI, whereas Model E yielded the best AIC score (28.20). Accordingly, we looked at other criteria to determine which model provided the optimal baseline on which to make comparisons of the three explanations. We chose Model D because it was the only one of the models that did not generate a warning that the measurement error matrix theta epsilon was positive definite. Nor did Model D have problems with q plots or the correlation of estimates. Model D was also in the middle of the range of competing models for both the NFI and AIC. Moreover, Model D's similar substantive parameters were constrained to equal each other in value, diagonal elements of psi were constrained, and theta epsilon was free. Model D had an excellent fit to the data with a chi-square value of 10.26, with 8 *df,* a probability level of .25, goodness of fit index of .96, an adjusted goodness of fit index of .90, a root mean square residual of .08, and a coefficient of determination of .998 for the *y* variables. The estimates of errors of measurement were all nonsignificant for the single item indicants, ranging from

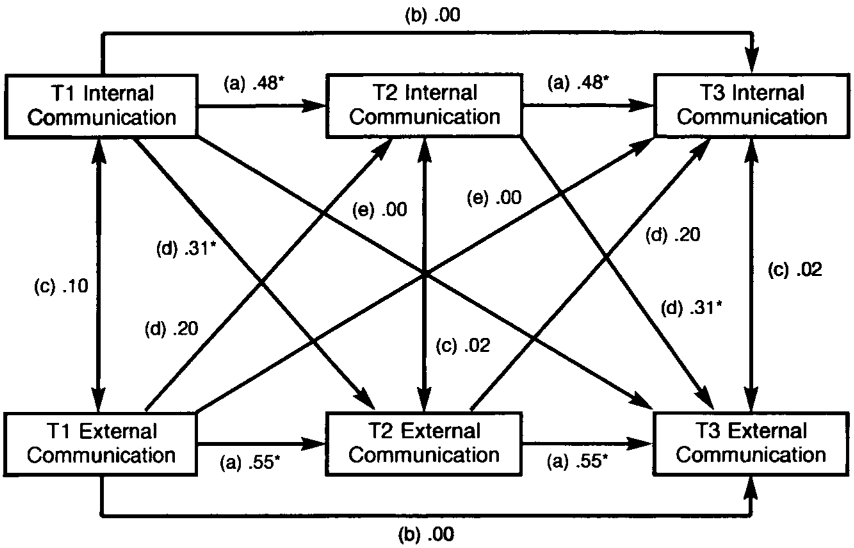
Table 2.
Comparison of Models of the Interrelationships between Internal and External Communication

Model	Substantive Interpretation	Chi-square	Df	NFI	AIC
Null	Baseline comparison	65.01	20	–	67.01
A	No constrained substantive parameters, theta epsilon fixed at 0.	7.70	4	.88	41.70
B	Substantive parameters constrained, theta epsilon fixed at 0.	11.43	9	.82	35.43
C	Substantive parameters constrained, on diagonal elements psi constrained, theta epsilon fixed at 0.	13.16	14	.80	37.16
D	Substantive parameters constrained, on diagonal lements psi constrained, theta epsilon free.*	10.26	8	.84	36.26
E	Substantive parameters constrained, on diagonal elements psi constrained, theta epsilon fixed constrained.*	12.20	13	.81	28.20
F	Some substantive parameters constrained, on diagonal elements psi constrained, theta epsilon fixed constrained.*	9.18	11	.86	29.18
G	Substantive parameters constrained, on diagonal elements psi constrained, theta epsilon free, lagged paths estimated.	5.99	4	.91	39.99
H	Substantive parameters constrained, on diagonal elements psi constrained, theta epsilon constrained, lagged paths constrained.	8.66	11	.87	28.66

*No identification problem with theta epsilon.

.22 for T1 internal to .48 for T3 internal communication. The diagonal elements of psi were all constrained to .51, a significant value, indicating a moderately high level of explained variance. The normalized residuals fell within acceptable ranges. The standard errors for the parameters were generally low to moderate. No obvious problems with identification

Figure 2.
Classic Panel Results for Model D of the Interrelationship between Internal and External Communication



*p < .05

T1, T2, and T3 = data-gathering periods

a = Stability at consecutive time points
b = Stability at lagged time points
c = Cross-sectional interrelationships
d = Consecutive interrelationships
e = Lagged interrelationships

and/or estimation were present in the results, and there was also no indication of colinearity in the estimates.

Comparing the Three Models

Figure 2 contains the parameter estimates for the substantive (beta) paths used to compare the models. The *a* paths were significant and high in value for both internal and external communication, suggesting a fair amount of stability in these variables over consecutive time points. The *b* paths were low and nonsignificant when estimated in Model E, indicating a mediating impact for intervening time points, which also provided some support for our measurement procedures. The *c* paths indicated little contemporaneous association between internal and external communication. The *d* paths were low to moderate in value, with the ones from internal communication to external communication significant. The *e* paths between internal and external indicated little lagged effect over two distant time periods.

In sum, the results for internal and external communication at consecutive time periods provided partial support for both H1A and B and H2A and B; that is, relationships between internal and external communication tended to remain stable over time, supporting both the functional specialization and the communication stars explanation. But there was no support for H1C, that internal and external communication would be negatively related, thus weakening the functional specialization explanation. Along these lines, partial support for H2C was found, suggesting a lagged effect for a communication stars explanation: that is, high levels of internal communication at the preceding time period producing high levels of lagged external communication. With the lone exception of H3F, positive relationships between internal and external communication at consecutive time periods, there was no support for a cyclical explanation.

Discussion

Model D provided an excellent fit to the data, with slightly more support for a communication stars explanation for the interrelationships between internal and external communication than for the other two explanations. In this section we discuss these findings as they pertain to the roles of internal and external communication in the operations of the CIS. Study limitations and implications for future research are also presented.

External Communication and the CIS

Finding a significant path from internal to external communication, but not vice versa, at consecutive time periods for the *d* paths fits well with the CIS's plans for external communication. In addition to their traditional representational and gatekeeping functions, boundary spanners in the CIS also develop community coalitions as a way of building political support from various stakeholders for their ongoing innovation efforts. CIS boundary spanners also increase the reach and impact of the CIS in this manner, thereby taking a more proactive role in resource dependence issues (Mizruchi & Galaskiewicz, 1993).

The new OM position was intended to produce a 'multiplier effect' for NCI efforts by developing community coalitions designed to address various cancer-related issues (see also Kaluzny et al., 1993; Kaluzny & Warnecke, 1996). Thus, the CIS recognized a central tenet of organizations in competitive environments: They must seek cooperative relationships with other organizations (Kumar, Stern, & Anderson, 1993). One way this approach is "sold" to community coalitions is capacity building, or pooling resources from various community organizations to create a whole that is more than the sum of its parts. In this approach the CIS provides technical assistance (e.g., training, public education materials) in the creation of campaigns, serves as a facilitator for coalitions, and provides access to resources (Ballard, 1996).

It is also interesting to note that these external relationships flow from the CIS based on centralized, internal planning, with little explicit feedback from community organizations. OMs can act as change agents (in classic diffusion of innovation frameworks), who identify intermediaries to serve as opinion leaders for their groups (Rogers, 1983, 1995). In sum, OMs are typically more interested in developing communication campaigns and coalitions at the regional level, in part related to the CISRC Project 3 intervention strategy, suggested by the moderate, significant path between internal communication and external communication at subsequent time periods.

Communication Within CIS

As we have seen, CIS members appear to emphasize communication in their regions rather than boundary spanning across ROs within the CIS network (Johnson, Chang, Pobocik, Meyer, Ethington, Ruesch, Wooldridge, & Murphy, 1995). One early concern related to the operation of the CISRC internal intervention strategies was that they might detract, given the press of work and the finite resources of the CIS, from other intervention strategy efforts. However, CIS members appeared to be more involved in external intervention strategy efforts in their communities than internal ones. This finding may be attributable to CIS members' greater control, influence, sympathy, and understanding of these efforts (Chang, Johnson, Cox, & Kiyomiya, 1997). In general, other researchers have noted the relative scarceness of internal innovation communication, which, given its importance to innovation and keeping an organization on the cutting edge, enhances the importance of efforts designed to sustain and enhance it (Albrecht & Ropp, 1984). In the CIS there was a clear difference between internally and externally oriented innovation-related communication, with external communication much more frequent. This pattern of external communication is typical of organizations that want to maintain their relative autonomy within a community (Oliver, 1991).

In prior work we have identified a variety of other explanations for the relatively low levels of internal innovation-related communication: a minimalist explanation that suggests certain innovations do not require much communication to implement; a left-hand censoring explanation suggesting that buy-in occurred before the start of data collections; a need-to-know explanation suggesting that the dispersed implementation of the three projects left some people unaware of how the internal innovations were unfolding; a sampling interval explanation; and an explanation suggesting that that the real action in internal innovation communication was at national face-to-face meetings (Chang, Johnson, Cox, & Kiyomiya, 1997; Johnson, Bettinghaus, Woodworth, Fleisher, Ward, & Meyer, 1997; Pobocik, Johnson, Darrow, Muha, Thomsen, Steverson, Stengle, & Ward, 1997). Put simply, it may also be that planning an innovation is a less communication-intensive activity than is its implementation by means of a community-based coalition.

Limitations

Several limitations of the present study could account for our findings. First, while the present study employed measurement methods that have been frequently used in previous research (Marsden, 1990), estimates of reliability are difficult with only one measure. However, the relative uniformity of the *a* paths would suggest some stability in the single item measures used in this study. Alternatively, it could suggest a certain inertia in communication activities once roles are established. Second, the sample sizes of this organizational census were relatively small for LISREL analyses (Hayduk, 1989), although in this instance we did not evaluate a multi-measure, latent trait model. Third, it is generally preferable to have equal sampling intervals. However, the unequal intervals between data collection points (11 & 15 months, respectively) used in this study appeared not to have made a major difference in the paths across two time points, except that the *a* path estimates were substantially higher between T2 and T3 than for T1 and T2 for external communication. These data reflected the stabilizing of external communication after an initially high level of activity at T1. Moreover, this finding suggested an initial exploration of a range of community-based organizations by the relatively new OMs at T1, followed by a more steady, some might even say efficient, reliance on partners who have demonstrated their usefulness at T2 and T3 (Chang et al., 1997). However, it could also be the case that yearly intervals between data collections did not fully capture the development of cycles. Fourth, to maintain long-term relationships with respondents, we made changes in the instruments. With respect to the internal communication logs, a clarification of local versus national communication was made between T2 and T3 and corrected by removing intra-ROs contacts at T1 and T2. We also added two communication channels (i.e., facsimile and electronic mail) to the communication logs between T2 and T3 in order to capture a more complete picture of the CIS' communication activities. The increased use of other channels could also explain the considerable drop for internal interpersonal communication from T2 to T3. With respect to the external measurement instrument, in the interest of reducing the respondents' burden, we eliminated several categories of external organizations at T2 and T3 that were mentioned very sparingly at T1 (Johnson et al., 1997).

Conclusion

New organizational forms depend on external communication for dealing with ever more complex interorganizational relationships. Accordingly, future research should build on the present study, perhaps by developing even more precise and sophisticated views of interrelationships between internal and external innovation communication. For example, Fiol (1989) has suggested that the strength of boundaries at various levels is critical, with strong boundaries between internal units leading to greater participation in joint ventures. This study suggested that a communication stars explanation, with some mix of formal planning elements, provides the best explanation of the interrelationships between internal and external communication. More studies need to be conducted to determine the balance between formalized structure and emergent communication networks in individual patterns of internal and external communication in new organizational forms. Contrary to recent trends in the literature related to unitary organizations (Damanpour, 1991; Johnson, 1993), some structure, as represented by formalization and planning, must also be present, to reduce uncertainty in a new organizational form. In a related study in this research stream we found that formalization had positive direct and indirect impacts on innovativeness when examined in a system of other variables (Johnson, et al., 1997). Prior studies in this area have emphasized that boundary spanning is the mechanism that operationalizes environmental cues to the internal organizational structure (Corwin, 1972; Lozada & Calantone, 1996; Spekman, 1979), but this study also suggests that they can serve as the mechanism by which organizations reach out to communities in even broader coalitions to achieve the objectives of new organizational forms.

NOTES

1. At T1, CIS staff were being trained on the various intervention projects. During T2, Project 1 shifted its focus from proactive counseling for promoting mammography to promoting 5-a-day fruit and vegetable consumption. Project 2 began its follow-up studies for making outcalls to promote mammography and Project 3 began its tests calling for the smoking cessation campaign. At T3, the 5-a-day project began its training, while preliminary results from the other two projects were being discussed and evaluated. Thus, the CIS members might need to contact different groups of people as the innovation projects went through the stages of planning, implementation, and review. The differential participation of the 19 ROs in the various projects could have contributed to the unbalanced communication distribution and considerable variance over time: 12 offices were involved with Project 1, one office participated in Project 2, and four offices engaged in Project 3.

2. Considerable effort was expended during the first year of the project on the development of data-gathering instruments. Extensive pretests were conducted during the summer of 1993. These pretests resulted in substantial modifications to the communication logs. The original instruments developed in the grant proposal were reviewed and revised based on additional research on the nature of the CIS and further review of the literature. The criteria for evaluating the results of pretests following in rank order in importance from first to last: instruments should (a) be likely to result in the high response rates needed for successful network analysis (e.g., 95 percent); (b) minimize respondent burden; (c) be couched in terms that are readily understood by respondents; and (d) have compatible operationalizations across different methods of data collection.

With these criteria in mind, we considered several alternative formats of the communication log. For example, instead of a roster it was decided to use a combination of communication log and directory, which minimized respondent burden, while also reminding respondents of the composition of the network. Based on the initial pretesting, it was also decided to change the content categories from operational to work-related and from innovation to intervention strategies based on feedback from respondents. The operational category was unfamiliar to respondents and innovation was a constant in this information services organization. Other researchers have experienced similar difficulties with respondents making distinctions between production and innovation related contents (Bach, 1989; Cheney, Block, & Gordon, 1986) and still others have noted on a conceptual level problems in distinguishing innovation and production (Stohl & Redding, 1987).

Since the major focus of this project was evaluation of new intervention strategies designed to reach target audiences within the CIS, it was decided it would be more appropriate to focus on this more limited type of innovation, which also may clear up some of the confusion found in prior studies when the broader category of innovation was used. While the CIS traditionally has engaged in a number of specific types of campaigns designed to reach target audiences, this type of activity has often been sporadic and ad hoc, focusing on national initiatives. The CISRC program project was designed to gradually and systematically increase the adoption of specific intervention strategies within the CIS network. Accordingly, the intervention strategies category, while initially unfamiliar to some members of the network, would become increasingly familiar to them as the CISRC program project developed. Other work-related communication would provide an interesting baseline on which to compare the development of intervention strategy-related communication. Responses to open-ended questions concerning what operational and innovation messages meant to respondents were used to craft definitions and examples used in the next rounds of pretesting. It was also decided not to include other categories of communication (e.g., social) because of concerns over the sensitivity of respondents and respondent burden, since each additional content category vastly increases it (Marsden, 1990).

3. Historically, in the project there has been some confusion over whether to record local vs. national communication in the logs. After repeated concerns were addressed to us, we decided in the next data collection, May 1995, to make clear the very limited situations in which communication at a local level should be reported. To insure no confusion was present in the current study all intra-RO communication links were removed before analyses were conducted.

4. The content of relationships has generally presented a difficult problem in network analysis research with a variety of strategies developed to deal with this problem (Burt & Schott, 1985). Typically, a network analyst makes a tradeoff between simplicity at the dyadic level to examine complexity at the social system level. Thus, in this research, we isolated those contents most directly related to the operation of the CIS as a system. Researchers must also confront the problem of differential meaning between members of the study population and themselves (Burt & Schott, 1985). Since this is a relatively new concept within the CIS, it is expected that over time members of the network will converge on a common meaning for this content.

REFERENCES

Adams, J. S. (1976). The structure and dynamics of behavior in organizational boundary roles. In M. D. Dunnette (Ed.), *Handbook of industrial and organizational psychology* (pp. 1175-1199). Chicago: Rand McNally.

Albrecht, T. L., & Ropp, V. A. (1984). Communicating about innovation in networks of three U.S. organizations. *Journal of Communication, 34,* 78-91.

Aldrich, H., & Herker, D. (1977). Boundary spanning roles and organizational structure. *Academy of Management Review, 2,* 217-230.

Allen, M. W. (1989). *Factors influencing the power of a linking role: An investigation into interorganizational boundary spanning.* Unpublished doctoral dissertation. Baton Rouge, LA: Lousiana State University.

Anderson, J. C., & Gerbing, D. W. (1988). Structural equation modeling in practice: a review and recommended two-step approach. *Psychological Bulletin, 103,* 411-423.

Arnold, M. F., & Hink, D. L. (1968). Agency problems in planning for community health needs. *Medical Care, 6,* 454-466.

At-Twaijri, M. I. A., & Montanari, J. R. (1987). The impact of context and choice on the boundary-spanning process: An empirical extension. *Human Relations, 40,* 783-798.

Bach, B. (1989). The effect of multiplex relationships upon innovation adoption: A reconsideration of Rogers' model. *Communication Monographs, 56,* 133-150.

Baker, W. E. (1992). Network organization in theory and practice. In N. Nohria & R. G. Eccles (Eds.), *Networks and organizations: Structure, form, and action* (pp. 397-429). Boston, MA: Harvard Business School Press.

Ballard, D. (1996, November). *Capacity-building approach to breast cancer awareness and screening.* Presentation to the Michigan Department of Community Health, Lansing, MI.

Bentler, P. M., & Bonnett, D. G. (1980). Significance tests and goodness of fit in the analysis of covariance structures. *Psychological Bulletin, 88,* 588-606.

Berelson, B., & Steiner, G. A. (1964). *Human behavior: An inventory of scientific findings.* New York: Harcourt, Brace & World.

Blau, J. R., & Alba, R. D. (1982). Empowering nets of participation. *Administrative Science Quarterly, 27,* 363-379.

Bollen, K. A., & Long, J. S. (1993). Introduction. In K. A. Bollen & J. S. Long (Eds.), *Testing structural equation models* (pp. 1-9). Newbury Park, CA: Sage.

Browning, L. D., Beyer, J. M., & Shetler, J. C. (1995). Building cooperation in a competitive industry: SEMATECH and the semiconductor industry. *Academy of Management Journal, 38,* 112-151.

Burk, J. (1994). Training MNC employees as culturally sensitive boundary spanners. *Public Relations Quarterly, 39,* 40-44.

Burt, R. S., & Schott, T. (1985). Relation contents in multiple networks. *Social Science Research, 14,* 287-308.

Carroll, G. R., & Teo, A. C. (1996). On the social networks of managers. *Academy of Management Journal, 39,* 421-440.

Chang, H.-J., Johnson, J. D., Cox, D., & Kiyomiya, T. (1997). *The communication environment of the CISRC: External communication and boundary spanning.* Paper presented to the International Communication Annual Convention, Montreal, Canada.

Cheney, G., Block, B. L., & Gordon, B. S. (1986). Perceptions of innovativeness and communication about innovations: A study of three types of service organizations. *Communication Quarterly, 34,* 213-230.

Corwin, R. G. (1972). Strategies for organizational innovation an empirical comparison. *Administrative Science Quarterly, 37,* 441-454.

Czepiel, J. A. (1975). Patterns of interorganizational communications and the diffusion of a major technological innovation in a competitive industrial community. *Academy of Management Journal, 18,* 6-24.

Daft, R. L. (1978). A dual-core model of organizational innovation. *Academy of Management Journal, 21,* 193-210.

Damanpour, F. (1991). Organizational innovation: A meta-analysis of effects of determinants and moderators. *Academy of Management Journal, 34,* 555-590.

DeBresson, C., & Amesse, F. (1991). Networks of innovators: A review and introduction to the issue. *Research Policy, 20,* 363-379.

Dillman, D. A. (1978). *Mail and telephone survey: The total design method.* New York: John Wiley.

Dillman, D. A. (1991). The design and administration of mail surveys. *Annual Review of Sociology, 17,* 225-249.

Farace, R. V., Monge, P. R., Bettinghaus, E. P., Eisenberg, E. M., White, L., Kurchner-Hawkins, R., & Williams, K. I. (1982). *Communication and coordination among health care organizations: Experiences from the Metropolitan Detroit Cancer Control Program.* E. Lansing, Mich: Department of Communication, Michigan State University.

Finkel, S. E. (1995). *Causal analysis with panel data.* Thousand Oaks, CA: Sage.

Fiol, C. M. (1989). A semiotic analysis of corporate language: Organizational boundaries and joint venturing. *Administrative Science Quarterly, 34,* 277-303.

Friedman, R. A., & Podolny, J. (1992). Differentiation of boundary spanning roles: Labor negotiations and implications for role conflict. *Administrative Science Quarterly, 37,* 28-47.

Galbraith, J. R. (1974). Organization design: An information processing view. *Interfaces, 4,* 28-36.

Galbraith, J. R. (1995). *Designing organizations.* San Francisco: Jossey-Bass.

Ghosal, S., & Bartlett, C. A. (1987). *Innovation processes in multinational corporations.* Cambridge, MA: Harvard Business School Press.

Gibson, D. V., & Rogers, E. M. (1994). *R&D collaboration on trial.* Cambridge, MA: Harvard Business School Press.

Goes, J. B., & Park, S. H. (1997). Interorganizational links and innovation: The case of hospital services. *Academy of Management Journal, 40,* 673-696.

Grover, V., Jeong, S.-R., Kettinger, W. J., & Lee, C. C. (1993). The chief information officer: A study of managerial roles. *Journal of Management Information Systems, 10,* 107-130.

Hakansson, H., & Sharma, D. D. (1996). Strategic alliances in a network perspective. In D. Iacobucci (Ed.), *Networks in marketing* (pp. 108-124). Thousand Oaks, CA: Sage.

Hayduk, L. A. (1987). *Structural equation modeling with LISREL.* Baltimore: Johns Hopkins University Press.

Jemison, D. B. (1984). The importance of boundary spanning roles in strategic decision-making. *Journal of Management Studies, 21,* 131-152.

Johnson, J. D. (1993). *Organizational communication structure.* Norwood, NJ: Ablex.

Johnson, J. D., Berkowitz, J., Ethington, C., & Meyer, M. (1994a). *General appendices to network analysis technical reports.* East Lansing, MI: Department of Communication, Michigan State University.

Johnson, J. D., Berkowitz, J., Ethington, C., & Meyer, M. (1994b). *Technical report*

#1: Analysis of the CIS/Program Project Network for the period October 1993 to March 1994. E. Lansing, MI: Department of Communication, Michigan State University.

Johnson, J. D., Bettinghaus, E., Woodworth, M., Fleisher, L., Ward, J. A., & Meyer, M. (1997). *Lessons learned: Implications for theory and the practice of research on communication networks.* Paper presented to the International Communication Association Annual Convention, Montreal, Canada.

Johnson, J. D., Chang, H. J., Ethington, C., Meyer, M., & La France, B. H. (1994). *Technical report #2: Analysis of the CIS/Program Project Network for the period April 1994 to September 1994.* E. Lansing, MI: Department of Communication, Michigan State University.

Johnson, J. D., Chang, H. J., La France, B. H., Meyer, M., Ethington, C., & Pobocik, S. (1996a). *Network Analysis Technical Report #4: Analysis of the CIS/Program Project Network for the period March 1995 to June 1995.* E. Lansing, MI: Department of Communication, Michigan State University.

Johnson, J. D., Chang, H.-J., LaFrance, B., Meyer, M., Ethington, C., & Pobocik, S. (1996b). *Technical Report #6: Analysis of the CISRC Project Network for the period November 1995 to May 1996.* East Lansing, MI: Department of Communication, Michigan State University.

Johnson, J. D., Chang, H. J., Pobocik, S., Meyer, M., Ethington, C., Ruesch, D., Wooldridge, & Murphy, J. (1995). *Functional work groups and evaluation of communication channels: Comparisons of six competing theoretical perspectives.* Paper presented to the Organizational Communication Division at the Speech Communication Association Annual Convention, San Antonio, TX.

Johnson, J. D., LaFrance, B. H., Meyer, M., Speyer, J. B., & Cox, D. (1998). The impact of formalization, role conflict, role ambiguity, and communication quality on perceived organizational innovativeness in the Cancer Information Service. *Evaluation and the Health Professions, 21,* 27-51.

Johnson, J. D., Meyer, M., Berkowitz, J., Ethington, C., & Miller, V. (1997). Testing two contrasting models of innovativeness in a contractual network. *Human Communication Research, 24,* 320-348.

Kaluzny, A. D., Lacey, L. M., Warnecke, R., Hynes, D. M., Morrissey, J., Ford, L., & Sondik, E. (1993). Predicting the performance of a strategic alliance: An analysis of the Community Clinical Oncology Program. *Health Services Research, 28,* 159-182.

Kaluzny, A. D., & Warnecke, R. B. (1996). *Managing a health care alliance: Improving community cancer care.* San Francisco: Jossey-Bass.

Katz, E., & Lazersfeld, P. F. (1955). *Personal influence: The part played by people in the flow of mass communications.* New York: The Free Press.

Katz, R., & Tushman, M. (1979). Communication patterns, project performance, and task characteristics: An empirical evaluation and integration in an R&D setting. *Organizational Behavior and Human Performance, 23,* 139-162.

Katz, R., & Tushman, M. L. (1981). An investigation into the managerial roles and career paths of gatekeepers and project supervisors in a major R&D facility. *R&D Management, 11,* 103-110.

Keller, R. T., Szillagyi, A. D., & Holland, W. E. (1976). Boundary spanning activity and employee reactions: An empirical study. *Human Relations, 29,* 699-710.

Kessler, L., Fintro, C., Muha, C., Wun, L. M., Annett, D., & Mazen, K. D. (1993). The Cancer Information Service Telephone Evaluation and Reporting System

(CISTERS): A new tool for assessing quality assurance. *Journal of the National Cancer Institute Monographs, 14,* 61-66.

Kimberly, J. R. (1978). Hospital adoption of innovation: The role of integration into external informational environments. *Journal of Health and Social Behavior, 19,* 361-373.

Kotter, J. P. (1979). Managing external dependence. *Academy of Management Review, 4,* 87-92.

Kumar, N., Stern, L. W., & Anderson, J. C. (1993). Conducing interorganizational research using key informants. *Academy of Management Journal, 36,* 1633-1651.

Lauman, E. O., Marsden, P. V., & Prensky, D. (1983). The boundary specification problem in network analysis. In R. S. Burt & M. J. Minor (Eds.), *Applied network analysis* (pp. 18-34). Beverly Hills, CA: Sage.

Lozada, H. R., & Calantone, R. J. (1996). Scanning behavior and the process of organizational innovation. *Journal of Managerial Issues, 8,* 310-325.

Luke, R. D., Begun, J. W., & Pointer, D. D. (1989). Quasi-firms: Strategic interorganizational forms in the health care industry. *Academy of Management Review, 14,* 9-19.

Lysonski, S. J., & Johnson, E. M. (1983). The sales manager as a boundary spanner: A role theory analysis. *Journal of Personal Selling & Sales Management, 3,* 8-21.

Manev, I. M., & Stevenson, W. B. (1996). *Balancing ties: Internal and external contacts in the organization's extended network of communication.* Paper presented at annual meeting of the Academy of Management, Cincinnati, OH.

Marsden, P. V. (1990). Network data and measurement. *Annual Review of Sociology, 16,* 435-463.

Meyer, M., Johnson, J. D., & Ethington, C. (1997). Contrasting attributes of preventive health innovations. *Journal of Communication, 47,* 112-131.

Miles, R. E., Snow, C. C., Matthews, J. A., Miles, G., & Coleman, H. J. Jr. (1997). Organizing in the knowledge age: Anticipating the cellular firm. *Academy of Management Executive, 11,* 7-20.

Mintzberg, H. (1973). *The nature of managerial work.* New York: Harper & Row.

Mizruchi, M. S., & Galaskiewicz, J. (1993). Networks of interorganizational relations. *Sociological Methods & Research, 22,* 46-70.

Monge, P. R., & Eisenberg, E. M. (1987). Emergent communication networks. In C. R. Berger & S. H. Chaffee (Eds.), *Handbook of communication science* (pp. 239-270). Newbury Park, CA: Sage.

Morra, M., Bettinghaus, E. P., & Marcus, A. C. (1993). The first 15 years: What has been learned about the Cancer Information Service and the implications for the future. *Journal of the National Cancer Institute Monographs, 14,* 177-185.

Morra, M., Van Nevel, J.P., Nealon, E., Mazan, K. D., Thomsen, C. (1993). History of the Cancer Information Service. *Journal of the National Cancer Institute, 14,* 7-34.

Nagpaul, P. S., & Pruthi, S. (1979). Problem-solving and idea-generation in R&D: The role of informal communication. *R&D Management, 9,* 147-149.

Oliver, C. (1990). Determinants of interorganizational relationships: Integration and future directions. *Academy of Management Review, 15,* 241-265.

Oliver, C. (1991). Network relations and loss of organizational autonomy. *Human Relations, 44,* 943-961.

Paisley, W. (1980). Information and work. In B. Dervin & M. J. Voight (Eds.), *Progress in communication sciences, Volume 2* (pp. 114-165). Norwood, NJ: ABLEX.

Pobocik, S., Johnson, J. D., Darrow, S., Muha, C., Thomsen, C., Steverson, D., Stengle, W., & Ward, J. (1997). *Internal communication mechanisms within the Cancer Information Service.* Paper presented to the International Communication Association Annual Convention, Montreal, Canada.

Reynolds, E. V., & Johnson, J. D. (1982). Liaison emergence: Relating theoretical perspectives. *Academy of Management Review, 7,* 551-559.

Robertson, T. S., & Wind, Y. (1983). Organizational cosmopolitanism and innovativeness. *Academy of Management Journal, 26,* 332-338.

Rogers, E. M. (1983). *Diffusion of innovations* (3rd. ed.). New York: The Free Press.

Rogers, E. M. (1995). *Diffusion of innovations* (4th. Ed.). New York: The Free Press.

Romanelli, E. (1991). The evolution of new organizational forms. *Annual Review of Sociology, 17,* 79-103.

Schwab, R. C., Ungson, G. R., & Brown, W. B. (1985). Redefining the boundary spanning-environment relationship. *Journal of Management, 11,* 75-86.

Sheppard, B. H., & Tuchinsky, M. (1996). Interfirm relationships: A grammar of pairs. *Research in Organizational Behavior, 18,* 331-373.

Singh, J., Goolsby, J. R., & Rhoads, G. K. (1994). Behavioral and psychological consequences of boundary spanning burnout for customer service representatives. *Journal of Marketing Research, 31,* 558-569.

Spekman, R. E. (1979). Influence and information: An exploratory investigation of the boundary role person's basis of power. *Academy of Management Journal, 22,* 104-117.

SPSS (1993). *SPSS LISREL7 and PRELIS.* Chicago: Author.

Starbuck, W. H. (1976). Organizations and their environments. In M. D. Dunnette (Ed.), *Handbook of industrial and organizational psychology* (pp. 1069-1123). Chicago: Rand McNally.

Stevenson, W. B. (1990). Networks of interaction within organizations. *Social Science Research, 19,* 113-131.

Stevenson, W. B., Pearce, J. L., & Porter, L. W. (1985). The concept of "coalition" in organization theory and research. *Academy of Management Journal,* 10, 256-268.

Stohl, C., & Redding, W. C. (1987). Message and message exchange processes. In F. M. Jablin, L. L. Putnam, K. H. Roberts, & L. W. Porter (Eds.), *Handbook of organizational communication: An interdisciplinary perspective.* Newbury Park, CA: Sage.

Tanaka, J. S. (1993). Multifaceted conceptions of fit in structural equation models. In F. A. Bollen & J. S. Long (Eds.), *Testing structural equation models* (pp. 10-39). Newbury Park, CA: Sage.

Tushman, M. L., & Scanlan, T. J. (1981a). Boundary spanning individuals: Their role in information transfer and their antecedents. *Academy of Management Journal, 24,* 289-305.

Tushman, M. L., & Scanlan, T. J. (1981b). Characteristics and external orientations of boundary spanning individuals. *Academy of Management Journal, 24,* 83-98.

Weedman, J. (1992). Informal and formal channels in boundary-spanning communication. *Journal of the American Society for Information Science, 43,* 257-267.

Williams, L. J., & Podsakoff, P. M. (1989). Longitudinal field methods for studying reciprocal relationships in organizational behavior research: Toward improved causal analysis. *Research in organizational behavior, Vol. 11*, 247-292.

Zoch, L. M. (1993). *The boundary spanning role: A dissertation synopsis.* Paper presented at the International Communication Association Annual Convention.